Area & Volume Maze ANSWER KEY

Start top-left. Leave answers in terms of π (no units needed). Go through the door with your answer.

START

1 Find the area of the triangle.
base 10, height 7

70

3 Find the area.
rhombus
$d_1 = 10$, $d_2 = 6$

60

Find the volume.
cylinder
radius 5, height 4

400π

Find the surface area.
sphere
radius 7

35

40

30

15

$\frac{100\pi}{3}$

40π

49π

2 Find the area.
trapezoid
$b_1 = 6$, $b_2 = 10$, $h = 5$

80

4 Find the volume.
rectangular prism
4 by 5 by 6

148

Find the volume.
triangular prism
base area 12, length 9

216

Find the volume.
cone
radius 3, height 8

21

300

120

100

96π

1200

600

6 Find the volume.
cylinder
radius 3, height 10

150

5 Find the surface area.
cube
edge 5

125

9 Find the surface area.
cylinder
radius 4, height 6

80π

10 Find the volume.
square pyramid
base edge 10, height 12

90π

30π

25

288π

64π

360

400

7 Find the volume.
cone
radius 4, height 9

48π

8 Find the volume.
sphere
radius 6

144π

Find the area.
triangle
base 9, height 8

72

FINISH

You made it!

Area & Volume Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **35**	$A = \frac{1}{2}bh \rightarrow A = \frac{1}{2}(10)(7) = 35$
2	Answer: **40**	$A = \frac{1}{2}(b_1 + b_2)h \rightarrow A = \frac{1}{2}(16)(5) = 40$
3	Answer: **30**	$A = \frac{1}{2}d_1d_2 \rightarrow A = \frac{1}{2}(10)(6) = 30$
4	Answer: **120**	$V = lwh \rightarrow V = 4(5)(6) = 120$
5	Answer: **150**	$SA = 6s^2 \rightarrow SA = 6(25) = 150$
6	Answer: **90π**	$V = \pi r^2h \rightarrow V = \pi(9)(10) = 90\pi$
7	Answer: **48π**	$V = \frac{1}{3}\pi r^2h \rightarrow V = \frac{1}{3}\pi(16)(9) = 48\pi$
8	Answer: **288π**	$V = \frac{4}{3}\pi r^3 \rightarrow V = \frac{4}{3}\pi(216) = 288\pi$
9	Answer: **80π**	$SA = 2\pi r^2 + 2\pi rh \rightarrow SA = 32\pi + 48\pi = 80\pi$
10	Answer: **400**	$V = \frac{1}{3}Bh \rightarrow V = \frac{1}{3}(100)(12) = 400$